

## LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

**Section:** 14. Functions

### Lecture: 157. Multiple Returns

#### Section 1: Lecture Summary

This lecture explains how Python functions can return multiple values at once. While most functions return a single value, Python allows returning several results simultaneously using tuples. The instructor demonstrates this concept through two examples:

1. A function that takes three arguments and returns both their sum and product.
2. A function that calculates a student's total marks, average, and pass/fail grade from three input marks, and returns all three.

Both examples show how multiple values are returned as a tuple and printed. The concept is emphasized as useful for practical programming and interview scenarios.

#### Section 2: Key Concepts and Explanations

##### - Functions Returning Multiple Values

- By default, a Python function returns one value.
- Python functions can return multiple values by listing them separated by commas in the return statement.
- These multiple returned values are actually bundled into a tuple automatically by Python.
- Returned values can be of any type (integers, strings, collections, etc.).
- This feature allows grouping several related results from a function neatly.

##### - Tuple as a Return Type

- A tuple is an immutable ordered collection of items.
- When a function returns multiple values, it returns them as a tuple implicitly.
- This tuple can be unpacked by the caller if desired.

##### - Example Use Case: Sum and Product Calculation

- Function takes three numbers as input.
- Calculates sum and product internally.

- Returns these two values together.

#### - Example Use Case: Student Result Calculation

- Function takes marks for three subjects.

- Computes total marks (sum), average marks (total divided by three), and determines pass/fail based on average  $\geq 45$ .

- Returns total, average, and grade as a tuple.

### Section 3: Example Code and Use Cases

```
def fun(a, b, c):  
    sum = a + b + c  
    prod = a * b * c  
    return sum, prod
```

```
print(fun(5, 6, 7))
```

This function returns the sum and product of three numbers as a tuple. The print statement outputs `(18, 210)`.

```
def result(mrk1, mrk2, mrk3):  
    total = mrk1 + mrk2 + mrk3  
    average = total / 3  
    if average >= 45:  
        grade = 'Pass'  
    else:  
        grade = 'Fail'  
    return total, average, grade
```

```
print(result(55, 60, 70))
```

This function calculates total marks, average, and pass/fail grade for a student from three marks. It returns all three as a tuple. The output is `(185, 61.666666666666664, 'Pass')`.

### Section 4: Key Takeaways

- Python functions can return multiple values by separating them with commas in the return statement.

- Multiple returns are packaged as a tuple automatically.

- Returned values can be different types (numbers, strings, etc.).

- This feature is helpful to return related grouped data from a function succinctly.

- Practical examples include returning calculated values like sum/product or total/average/grade from a function.

- Knowing multiple return values is important for Python programming proficiency and useful for interviews or certifications.